UNIVERSIDAD PERUANA CAYETANO HEREDIA

Workshop 01

Databases and Sequence Alignment

- **Databases**

In recent decades, large amounts of data regarding DNA, RNA, and protein sequences have been dilucidated. These sequences are stored in databases or digital repositories; the records of these databases and many others can be found at the National Center for Biotechnology Information (NCBI), part of the National Institute of Health (NIH), the biomedical research agency of the United States of America.

Each database has characteristics according to the type of data they store:

- **GenBank** (https://www.ncbi.nlm.nih.gov/genbank/) is the NIH's database of gene sequences. Contains a collection of annotated DNA and RNA sequences.
- **Refseq** (https://www.ncbi.nlm.nih.gov/refseq/) contains sets of sequences that are used as references, since they collect non-redundant and integral sequence records. This means that several reports on the same gene are considered, but at the same time these are processed to unite several similar sequences in a single record.
- **UniProt** (https://www.uniprot.org) contains protein sequences and their annotations into two categories: Records that have been manually curated (reviewed and evaluated) are stored in the Swiss-Prot database, while sequences that have been computationally analyzed (and are pending manual review) are stored in the TrEMBL database.
- **Protein Data Bank (PDB)** (https://www.rcsb.org) stores protein conformational data obtained from experiments that determine their 3D structure, such as: X-Ray crystallography, Nuclear Magnetic Resonance (NMR), electron microscopy (Cryo-EM), molecular modeling, among other techniques.
- **PubMed search engine** (https://pubmed.ncbi.nlm.nih.gov) which organizes bibliographic references to a wide variety of works in life sciences and biomedicine. It contains references to original articles, reviews, protocols, and book excerpts; however, PubMed is essentially a citation database, where only the abstract of academic texts is usually found, while PubMed Central database contains the complete scientific texts.

Glossary:

- **Sequence annotation:** Refers to the identification of elements in a DNA sequence and their description. Examples of elements found in a sequence are the coding sequence of a gene (CDS), regulatory regions, introns, etc. The descriptions of these include data such as the position of the elements, the function of the genes in the sequence and their comparison with other genomes.
- **Sequence Representation:** FASTA is the most common format for representing nucleotide or peptide sequences. This format consists of a first line beginning with the symbol ">" followed by the description of the sequence and the lines below correspond to the sequence of base pairs or amino acids represented in the IUB/IUPAC nomenclature. Details about this format and some exceptions can be found at the following link: https://zhanggroup.org/FASTA/.
- **Sequence identifiers:** The sequences registered in each database own an identifier called accession number. The accession numbers of the reference sequences (in RefSeq) are easy to identify since they usually start with two letters followed by an underscore "_". The following are some of the prefixes' descriptions used in RefSeq:

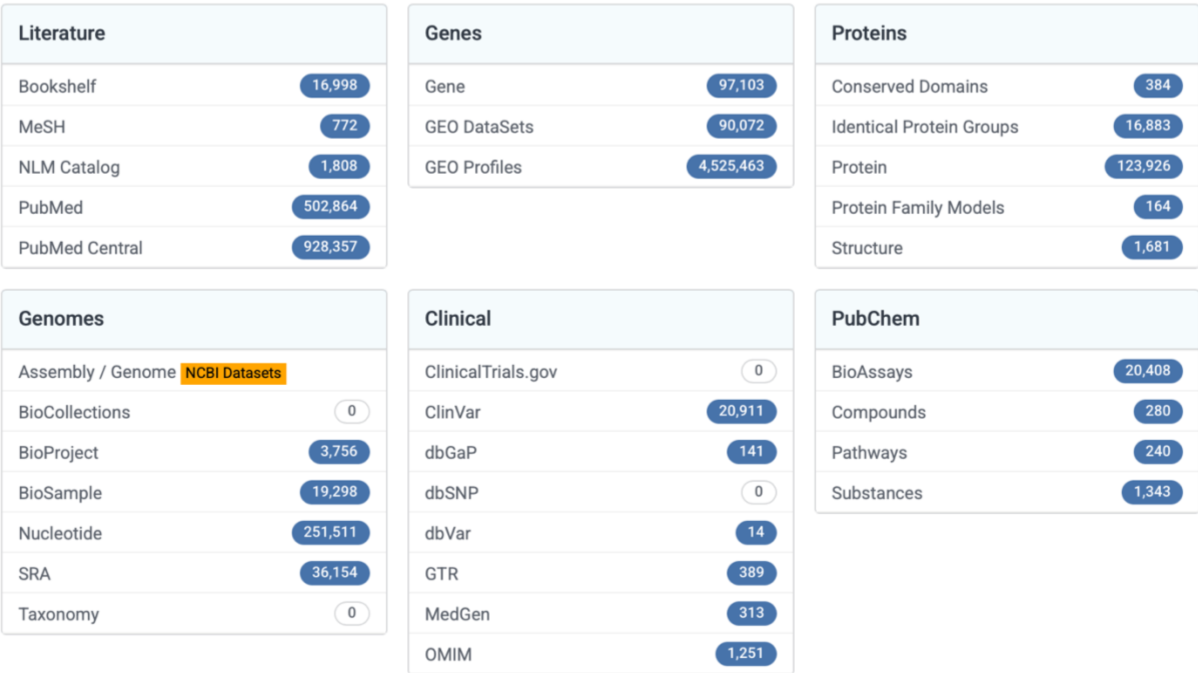
Accession Prefix	Molecule Type	Comment
AC_	Genomic	Complete genomic molecule, usually alternate assembly
NC_	Genomic	Complete genomic molecule, usually reference assembly
NG_	Genomic	Incomplete genomic region
NT_ [a]	Genomic	Contig or scaffold, clone-based or WGS
NW_ [a]	Genomic	Contig or scaffold, primarily WGS
NZ_ [b]	Genomic	Complete genomes and unfinished WGS data
NM_	mRNA	Protein-coding transcripts (usually curated)
NR_	RNA	Non-protein-coding transcripts
XM_ [c]	mRNA	Predicted model protein-coding transcript
XR_ [c]	RNA	Predicted model non-protein-coding transcript
AP_	Protein	Annotated on AC_ alternate assembly
NP_	Protein	Associated with an NM_ or NC_ accession
YP_ [c]	Protein	Annotated on genomic molecules without an instantiated transcript record
XP_ [c]	Protein	Predicted model, associated with an XM_ accession
WP_	Protein	Non-redundant across multiple strains and species

a. Whole Genome Shotgun sequence data.
b. An ordered collection of WGS sequence for a genome.
c. Computed.

Exploring NCBI:

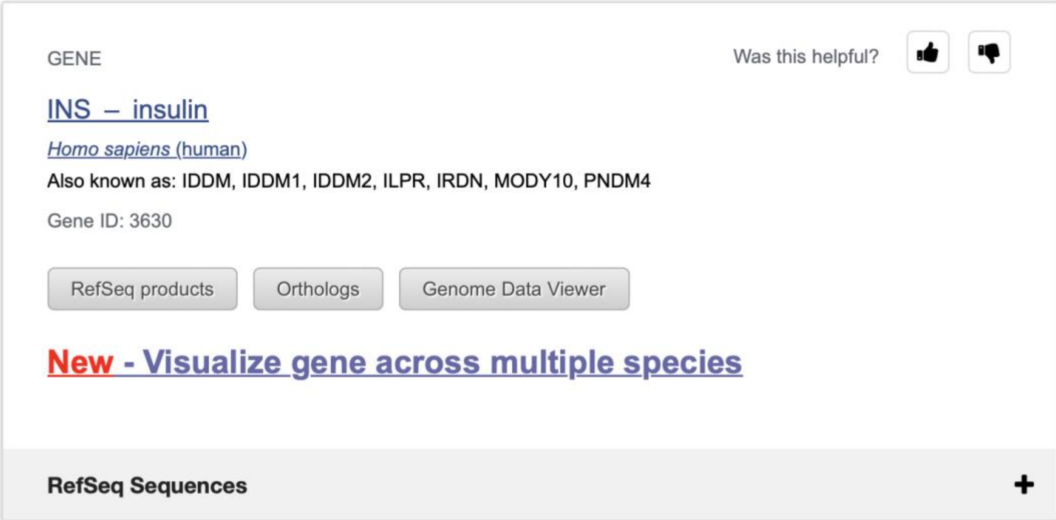
1. We entered the NCBI portal: https://www.ncbi.nlm.nih.gov/. The main page displays a search bar, where all available databases are searched by default.

2. It is recommended to specify the database where the search shall be performed. If you don't, then you will be prompted to select the database after attempting to search for the term. For example, if I enter the term **"insulin"**, I will get the following output:



3. We'll be searching for the insulin gene, so we should select the "Gene" database (or "Nucleotide" as well, because we want to know the sequence). It can be noticed that the number of results in "Gene" is very large (93 447 results). The reason is that the search has not been narrowed; the most effective way to do it is to specify the species of interest accompanying the scientific name with the [orgn] or [organism] command:

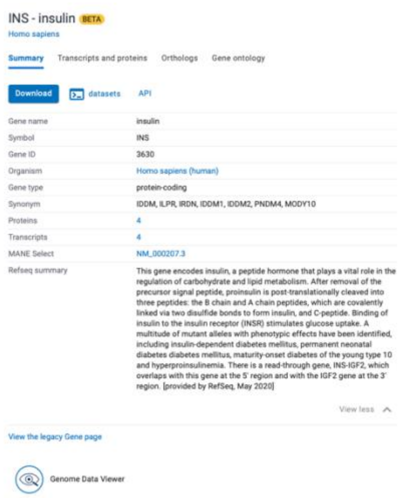
4. The website may show us a main result (secondary results might appear that could be genes that include the term "insulin" such as "insulin growth factor"). presented like this:



5. This expandable section shows RefSeq sequences that would correspond to the reference transcript and protein sequences of the gene. We could access this information directly, but first let's look at the information from the main result.

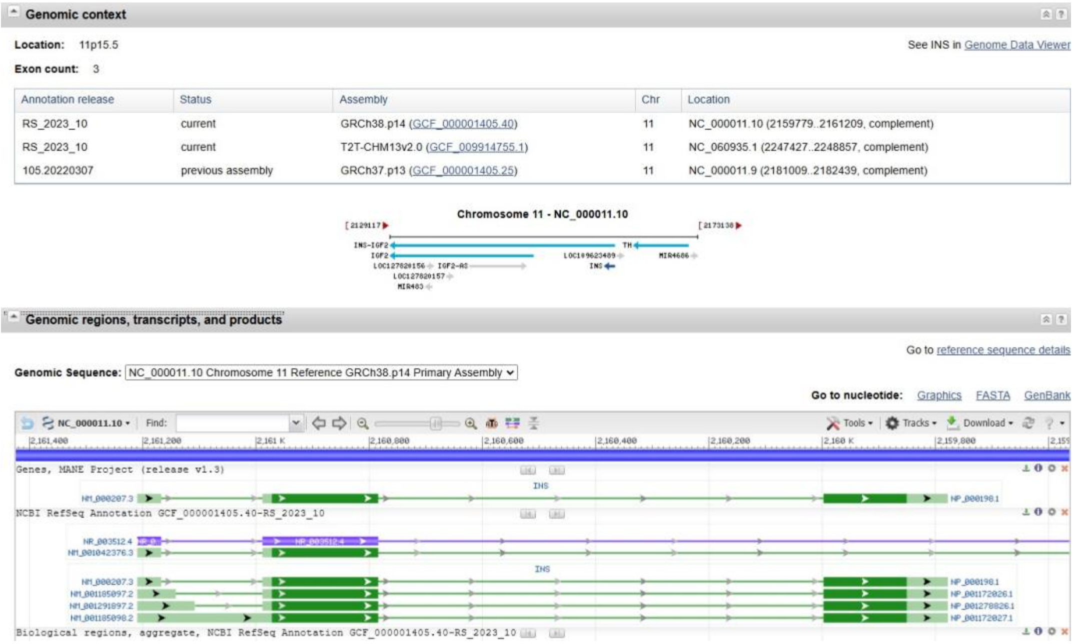
Transcript	nt	Protein	aa	Isoform	Status
NM_000207.3	465	NP_000198.1	110		MANE Select
NM_001185097.2	491	NP_001172026.1	110		curated
NM_001185098.2	644	NP_001172027.1	110		curated
NM_001291897.2	525	NP_001278826.1	110		curated

6. To get the full gene information select '**View the legacy Gene page'.** Many sections are displayed with extensive and valuable information of the gene. Nevertheless, we will focus on evaluating the sequence of the gene.

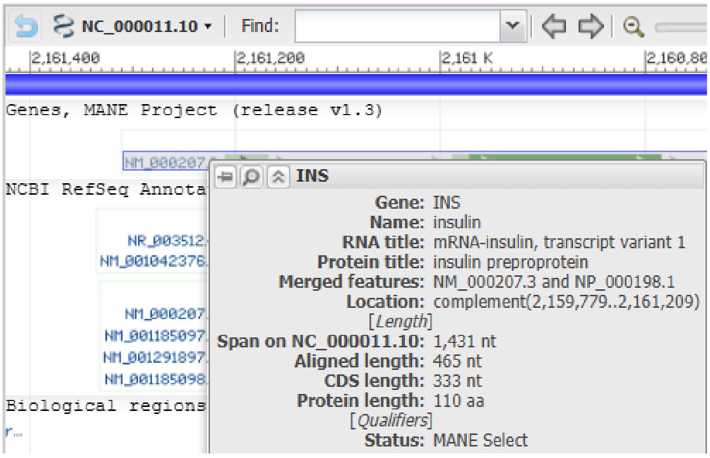


- The "Genomic context" section has information about the physical location of the gene within the genome. The result table indicates that the human insulin gene is located on chromosome 11, between nucleotides 2159779 and 2161209. And it also indicates that the gene has 3 exons.

- The "Genomic regions, transcripts, and products" section shows a genomic map in which the gene sequence is located.

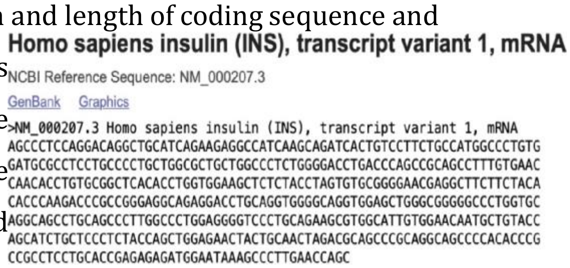


7. If we hover the cursor over each bar associated with an accession number, we will see that it indicates the gene that is represented. In this case, the insulin gene (INS) is represented by the MANE project accession number and six additional ones below.



However, we will use the one associated with the MANE Project (NCBI-EMBL-EBI Collaborative Gene Annotation Project).

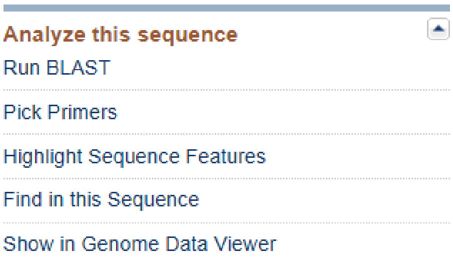
8. The general information of the gene (name, location and length of coding sequence and protein) can be observed. Below are other options that allow us to get more data, among these, the option to download the FASTA sequence of both the mRNA transcript and the protein. We will download these files.



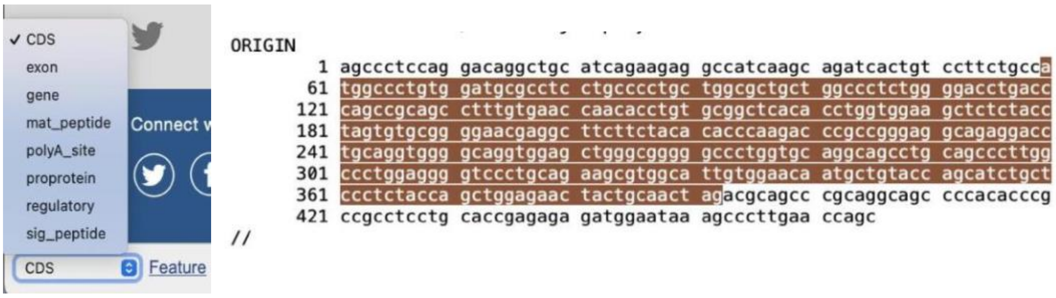
9. Other option allow us to view the gene element annotations registered in GenBank, by clicking the link NM_000207.3. Inhere besides the general description of the gene, it is important to recognize the FEATURES section that describes the elements of the sequence. Each element is represented by categories such as gene, exon, intron, coding sequence (CDS), mature peptide (mat_peptide), polyA site (polyA_site), proprotein, regulatory region (regulatory) and signal peptide (sig_peptide). On the right, the location of the element, the gene to which it belongs, and other descriptions such as the specific name of the element are declared. In the case of the CDS, the sequence of the translated product will be indicated.

```
CDS             60..392
                /gene="INS"
                /gene_synonym="IDDM; IDDM1; IDDM2; ILPR; IRDN; MODY10;
                PNDM4"
                /note="proinsulin; preproinsulin"
                /codon_start=1
                /product="insulin preproprotein"
                /protein_id="NP_000198.1"
                /db_xref="CCDS:CCDS7729.1"
                /db_xref="GeneID:3630"
                /db_xref="HGNC:HGNC:6081"
                /db_xref="MIM:176730"
                /translation="MALWMRLLPLLALLALWGPDPAAAFVNQHLCGSHLVEALYLVCG
                ERGFFYTPKTRREAEDLQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSICSL
                YQLENYCN"
```

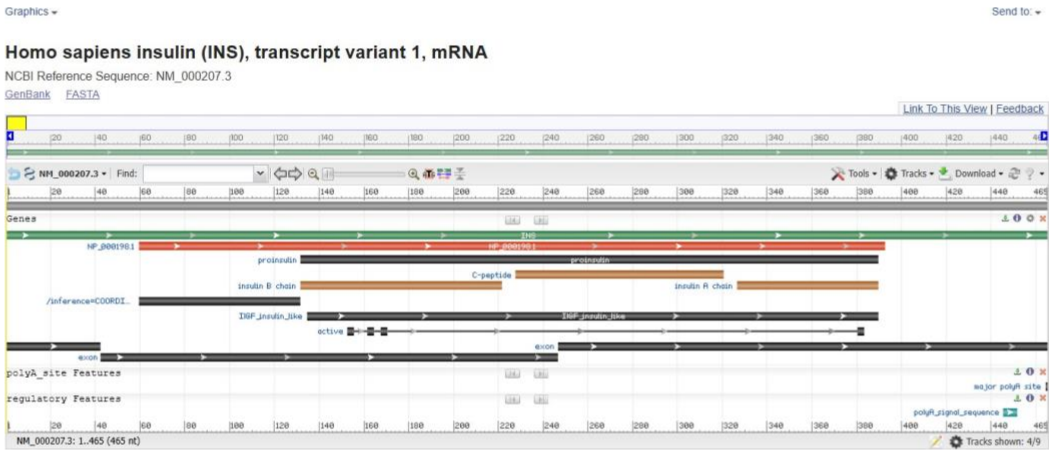
10. These elements can be found in an interactive way through a tool located in the upper right section of the website. The section is titled "Analyze this sequence". We click on the "Highlight Sequence Features" option and a toolbar will appear in the lower part of the window.



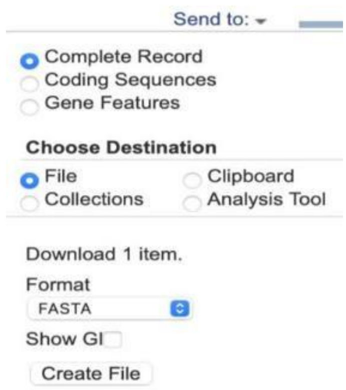
11. Selecting an item will highlight it in the sequence. In the example the coding sequence (CDS) was selected.



12. You can also browse the genetic map, for that, access the “Graphics” option at the top of the browser. There you can see graphically the location of some elements such as exons and mature peptides (insulin A and B chains).



You can download the displayed information by clicking on the “Send to” option, located at the top right of the sequence name. You can select what you want to download. If the option “Coding Sequences” is indicated, you will automatically be asked to choose between the FASTA format of the sequence in nucleotides or in amino acids.



Exploring UniProt:

1. The UniProt website is: https://www.uniprot.org/.

2. We'll look for the epidermal growth factor receptor (EGFR). Again, many results are returned because the search is not narrowed down by species. We can limit the species to *Homo sapiens* in the toolbar on the left (Human).

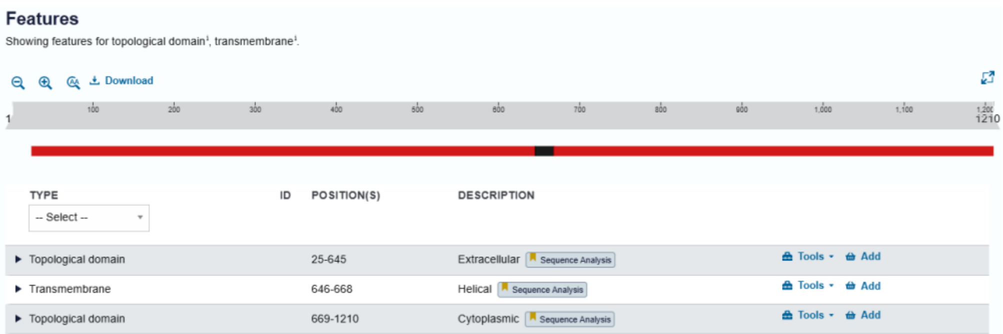
3. If our organism does not appear among the popular ones, we must specify it in the search bar highlighted in the figure above. Results show UNIPROT accession number, protein name, gene name, organism, and protein size (amino acids).

UniProtKB 344 results

Entry	Entry Name	Protein Names	Gene Names	Organism	Length
P01133	EGF_HUMAN	Pro-epidermal growth factor[...]	EGF	Homo sapiens (Human)	1,207 AA
Q12929	EPS8_HUMAN	Epidermal growth factor receptor kinase substrate 8	EPS8	Homo sapiens (Human)	822 AA
Q14451	GRB7_HUMAN	Growth factor receptor-bound protein 7[...]	GRB7	Homo sapiens (Human)	532 AA
P00533	EGFR_HUMAN	Epidermal growth factor receptor[...]	EGFR, ERBB, ERBB1, HER1	Homo sapiens (Human)	1,210 AA

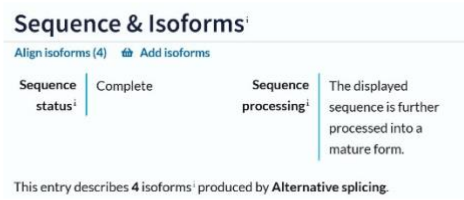
4. Like NCBI, there is a lot of valuable information reported in this database, however, given our interest in sequences, we will turn to 3 sections: “Subcellular location”, “Family & Domains” and “Sequences”.

   - "Subcellular location" section contains information regarding the cellular location of the protein. The protein can be in the cytoplasm, in the cell membrane or in some organelle. In the case of membrane proteins, the regions of a protein that span the cell membrane (transmembrane regions) and the extra and intracellular regions are indicated.



   - “Family & Domains” section is important to recognize which protein domains are directly involved in its function. For example, in the case of EGFR, its domain with kinase activity is located between amino acids 712 and 979.

- "Sequences & Isoforms" section contains the protein sequence. Several sequences can be found due to the existence of isoforms of the protein in question.

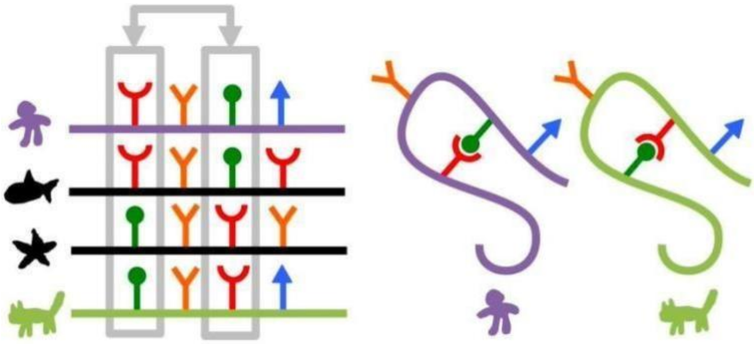


5. Each sequence can be downloaded individually by clicking on the download icon. The sequence will appear in a simple FASTA format in your browser. This can be downloaded by copying the sequence into a text file and saving it as the file name with the extension FASTA (.fasta) or .txt. You can also directly select 'copy sequence' and place it in the file of your choice. <u>Download the sequences of isoforms 1 and 2.</u>

Sequence Alignment

Aligning sequences is a widely used strategy that allows extracting useful information to formulate conjectures about the functional identity of a biomolecule based on the variations it presents. The purposes of the alignment are diverse and respond to the nature of the sequence with which the analysis begins (DNA/RNA or proteins).

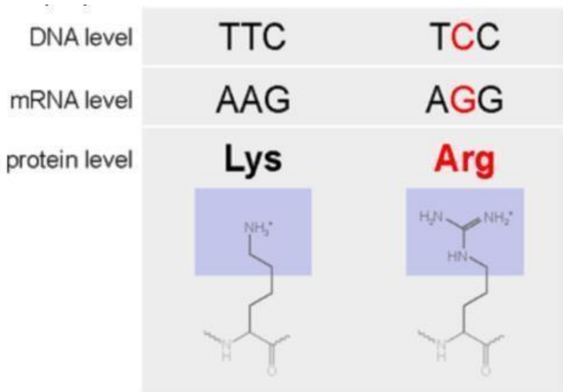
Through an alignment between sequences, it is possible, for example, to identify which regions of a protein have been subject to natural selection and/or are evolutionarily conserved between organisms. If we start from an unpublished sequence with an unknown function, through an alignment it is also possible to determine the degree of similarity that justifies the inference of a structural homology with another sequence with a known function.



Secuencia similar → Estructura similar → Función similar

In any circumstance, the variations between sequences (mutations) are informative elements that give us clues about the changes that a biomolecule has undergone throughout its evolutionary history and allow us to relate it to sequences that follow similar patterns. For example: By finding a high degree of sequence similarity between two proteins, we might infer that they share a common evolutionary history, and from this we might anticipate that they will have similar three-dimensional structures as well as similar biological functions. All this considering that the environmental conditions remain identical, so that the function has been preserved during evolution.

If we remember the versatility of the genetic code, we can anticipate that the effect of a mutation (even those that replace a single nucleotide) could be critical: changing a single nucleotide can mean that the amino acid that is translated from a triplet has very different physicochemical properties compared to those expected from the original sequence.

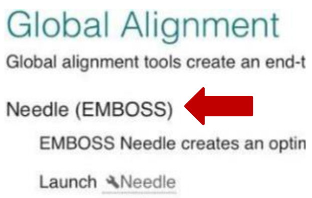


Global Alignment:

The premise of global alignment is to align each and every one of the residues of a sequence over the entire length when comparing it with another (protein or nucleotide sequences). By using the **Needleman–Wunsch algorithm**, a substitution matrix will be generated, and the quality of the aligned sequences will receive scores and penalties as the comparison is made.

The EMBL-EBI portal offers a variety of online bioinformatic tools; among these, the **Needleman–Wunsch algorithm** can be run from the EMBOSS Needle tool.

1. Enter the EMBL-EBI Tools > Pairwise Sequence Alignment portal available: https://www.ebi.ac.uk/Tools/psa/. Upon entering we go to the global alignment tools and choose to run the EMBOSS Needle algorithm.

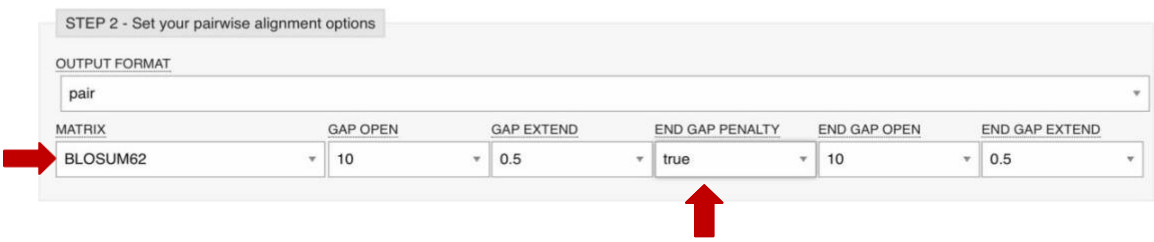


2. In this example, two protein sequences of prokaryotic serine-proteases (both recovered from UniProt) will be compared. The first (P29600) is the heat stable protease from *Bacillus lentus* commercialized under the trade name "Savinase". The second (P41363) is also a heat stable serine protease, but from a different species (*Bacillus halodurans*).

```
>P29600
AQSVPWGISRVQAPAAHNRGLTGSGVKVAVLDTGISTHPDLNIRGGASFVPGEPSTQDGNGHGTHVAGTIAALNNSIGVL
GVAPSAELYAVKVLGASGSGSVSSIAQGLEWAGNNGMHVANLSLGSPSPSATLEQAVNSATSRGVLVVAASGNSGAGSIS
YPARYANAMAVGATDQNNNRASFSQYGAGLDIVAPGVNVQSTYPGSTYASLNGTSMATPHVAGAAALVKQKNPSWSN
VQI RNHLKNTATSLGSTNLYGSGLVNAEAATR

>P41363
MRQSLKVMVLSTVALLFMANPAAASEEKKEYLIVVEPEEVSAQSVEESYDVDVIHEFEEIPVIHAELTKKELKKLKKDPNVKAI
EKNAEVTISQTVPWGISFINTQQAHNRGIFGNGARVAVLDTGIASHPDLRIAGGASFISSEPSYHDNNGHGTHVAGTIAALN
NSIGVLGVAPSADLYAVKVLDRNGSGSLASVAQGIEWAINNNMHIINMSLGSTSGSSTLELAVNRANNAGILLVGAAGNTG
RQGVNYPARYSGVMAVAAVDQNGQRASFSTYGPEIEISAPGVNVNSTYTGNRYVSLSGTSMATPHVAGVAALVKSRYPSY
TNNQIRQRINQTATYLGSPSLYGNGLVHAGRATQ
```

3. We need to adjust some parameters for the algorithm to be useful for our purposes. In the “More options” tab, let's make sure that the substitution matrix is **BLOSUM62** and that a penalty is considered for the gaps present at the extremes (end gap penalty > True). This way we will ensure that the gaps located at the end of the sequences also have an impact on the score. Then we proceed to execute the analysis by clicking on the Submit button.



4. Let's analyze the results screen and note that the degree of similarity between the aligned amino acids is defined by symbols that have the following convention:

```
AQL
||| Coincidence between amino acids is perfect
AQL

VLK Close mismatch.  value in the BLOSUM matrix is positive
 :::   Amino acids involved share physicochemical characteristics
IIR

RVN Far mismatch: There is no similarity between involved amino acids
...  value in the BLOSUM matrix is zero or negative
FAR
```

5. On the report screen it is possible to identify useful information to begin interpreting the alignment. For example, it is possible to check the algorithm that has been executed in the description of #Program and check the number of sequences that have been included in the analysis in the description of #Aligned_sequences. Likewise, it is possible to identify some parameters with which the algorithm has been executed, such as the substitution matrix #Matrix, the penalties for gaps in the interior #Gap_penalty and in the extremes #Extend_penalty and the length of the alignment #Length.

```
# Program: needle
# Rundate: Wed 17 Aug 2022 03:11:02
# Commandline: needle
#    -auto
#    -stdout
#    -asequence emboss_needle-I20220817-031332-0815-3905156-p1m.asequence
#    -bsequence emboss_needle-I20220817-031332-0815-3905156-p1m.bsequence
#    -datafile EBLOSUM62
#    -gapopen 10.0
#    -gapextend 0.5
#    -endopen 10.0
#    -endextend 0.5
#    -aformat3 pair
#    -sprotein1
#    -sprotein2
# Align_format: pair
# Report_file: stdout
########################################

#=======================================
#
# Aligned_sequences: 2
# 1: P29600
# 2: P41363
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 361
# Identity:     176/361 (48.8%)
# Similarity:   214/361 (59.3%)
# Gaps:          92/361 (25.5%)
# Score: 916.0
```

Finally, the screen shows us the main results of the alignment that provide useful information to make decisions about how related the sequences are. It includes percentages of #Identity, #Similarity, #Gaps and finally the score received by the alignment #Score.

Local Alignment:

Unlike a global alignment, a local alignment is used to identify similar regions that could indicate functional, structural, or evolutionary relationships between two protein or nucleotide sequences. This approach is certainly more useful when you have two sequences that are very different (composition or size) but are suspected of sharing regions or functional motifs. Consistent with this, the **Smith–Waterman algorithm** that describes this approach will prioritize matching between segments of all possible lengths in its substitution matrices, optimizing the similarity measure instead of looking for "fully" matches.

1. We re-enter the EMBL-EBI Tools > Pairwise Sequence Alignment portal available https://www.ebi.ac.uk/Tools/psa/. Upon entering we go to the local alignment tools and choose to run the **Water (EMBOSS) algorithm**.

2. In this example, similarity of two proteins will be compared. Both proteins (Tsol18 and Tso45) from the activated oncosphere of the *Taenia solium* parasite. They are currently being used as vaccines for swine cysticercosis. In this occasion, we will use the mRNA (cDNA) sequences available for these elements as input.

```
>TSOL18

GACGTTCACGACGACGAAGATGGTGTGTCGGTTTGCTCTCATCTTCTTGGTGGCCGTCGTTTTGGCGAGCGGTGACCGAACATTCGG
CGACGATATTTTCGTGCCATACCTTCGCTGCTTCGCCCTTAGCGCTACCGAAATTGGGGTGTTTTGGGATGCTGGAGAGATGGTTGG
CCATGGCGTAGAGGAGATCAAAGTGAAAGTAGAAAAAGCAATACACCCATACAAGATCTGGAATGCAACAGTCAGCGCGAACAAT
GGAAAAGTCATCATCAGAGACTTGAAGGCGAAGACAATTTACAGAGTGGACGTAGACGGTTATCGAAACGAAATCATGGTGTTTGG
TTCGCAGCGTTTCGCGACAACACTTCCGAAAAAGCAGATCAAGCACAAGAAGGTCCGAAGATCGTAGACGCTAGCATCGAACCTAA
AGTTTGCGCGGAGCAGGCAATGCACCTGTCGTCAATCATTTGCAAGTAAACAAGTATAATGAGACCTTTCATGTTAACGAATAAATC
TTCGATTACAAATACAGGGTGTTAGTAATGCTAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

>TSO45

ATGGCGTCTCAGTTCCACTTGATTCTATTGCTGACTTCAATTTTGGCTGGAAACCACAAGGCAACATCCCGCCATGGCGGAGTAACA
GTCGTGACGACTAGTGGATCTGGTATAGCCTCCGCAATACTTGGACTCCTCTTCACTTGCACGGTGCTAGTCCTCGCTTGAACACTCG
CCT
```

3. In the "More options" ta, let's make sure that the substitution matrix is **DNAfull**, then we proceed to execute the alignment with Submit. As in the previous section, the results screen will allow us to extract information to form conclusions.

```
# Program: water
# Rundate: Wed 17 Aug 2022 07:37:09
# Commandline: water
#    -auto
#    -stdout
#    -asequence emboss_water-I20220817-074416-0341-74653801-p2m.asequence
#    -bsequence emboss_water-I20220817-074416-0341-74653801-p2m.bsequence
#    -datafile EDNAFULL
#    -gapopen 10.0
#    -gapextend 0.5
#    -aformat3 pair
#    -snucleotide1
#    -snucleotide2
# Align_format: pair
# Report_file: stdout
########################################

#=======================================
#
# Aligned_sequences: 2
# 1: TSOL18
# 2: TSO45
# Matrix: EDNAFULL
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 369
# Identity:     123/369 (33.3%)
# Similarity:   123/369 (33.3%)
# Gaps:         215/369 (58.3%)
# Score: 174.5
#
```

Multiple alignment:

Lastly, the multiple sequence alignment (MSA) is an alignment of three or more biological sequences (DNA/RNA or proteins). In general, it is assumed that the set of sequences entered as input (problem set) have an evolutionary relationship by which they share a lineage and descend from a common ancestor. From the resulting MSA, homology can be inferred, and phylogenetic analysis can be carried out to assess the shared evolutionary origins of the sequences. MSA is often used to assess the conservation of protein domains, tertiary and secondary structures, and even individual amino acids or nucleotides.

Unlike global optimization algorithms, which also allow the analysis of more than two sequences, an MSA assumes the challenge of finding matches considering the set of sequences as a single element. Starting from this premise, MSAs require more sophisticated methodologies than pairwise alignment because they are more complex from a computational point of view. Most of the MSA algorithms use heuristic methods to achieve similarities.

1. We re-enter the EMBL-EBI portal this time to explore the MSA Tools > Multiple Sequence Alignment tools which are available at: http://www.ebi.ac.uk/Tools/msa/. Upon entering we go to the available tools and choose to run the **Clustal Omega tool**.

2. For this exercise, we will search the corresponding database for the amino acid sequences that correspond to the alpha subunit of hemoglobin from 5 different organisms: *Homo sapiens* (human), *Danio renio* (zebra fish), *Mus musculus* (mouse), *Capra hircus* (domestic goat) and *Pan troglodytes* (Chimpanzee). To enter the sequences, we will have to create a multi-FASTA file or directly copy and paste the FASTA files of each organism in the field.

```
>Pan troglodytes
MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAQVKGHGKKVADALTNAVA
HVDDMPNALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYR

>Danio renio
MSLSDTDKAVVKAIWAKISPKADEIGAEALARMLTVYPQTKTYFSHWADLSPGSGPVKKHGKTIMGAVGEAI
SKIDDLVGGLAALSELHAFKLRVDPANFKILSHNVIVVIAMLFPADFTPEVHVSVDKFFNNLALALSEKYR

>Homo sapiens
MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAQVKGHGKKVADALTNAVA
HVDDMPNALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYR

>Capra hircus
MSLTRTERTIILSLWSKISTQADVIGTETLERLFSCYPQAKTYFPHFDLHSGSAQLRAHGSKVVAAVGDAVK
SIDNVTSALSKLSELHAYVLRVDPVNFKFLSHCLLVTLASHFPADFTADAHAAWDKFLSIVSGVLTEKYR

>Mus musculus
MVLSGEDKSNIKAAWGKIGGHGAEYGAEALERMFASFPTTKTYFPHFDVSHGSAQVKGHGKKVADALANAAG
HLDDLPGALSALSDLHAHKLRVDPVNFKLLSHCLLVTLASHHPADFTPAVHASLDKFLASVSTVLTSKYR
```

3. With the selected sequences we proceed to perform the multiple alignment maintaining the default parameters and click on Submit. Let's analyze the results screen and note that the degree of similarity between the aligned amino acids is defined by symbols that have a similar convention to those of simple alignment:

```
AQL Los asteriscos indican que La coincidencia entre
AQL aminoácidos es perfecta: un residuo completamente
AQL conservado.
***

DKT  Indica  la  existencia  de  conservación  fisicoquímica
DKS entre grupos de residuos de propiedades similares.
ERT Scoring > 0.5 en la matriz Gonnet PAM 250.
:::

GSG Indica que el grado de conservación es débil o
GSG muestra  una  similitud  fisicoquímica  muy  débil.
SCA Scoring =< 0.5 en la matriz Gonnet PAM 250.
...
```

4. The results screen offers some options that will allow us to explore information to generate conjectures. To begin with, a visually informative way of being able to identify matches that exist in this set of sequences is to color-code the residuals. The “Show colors” button will allow you to explore the results following this guideline.

```
CLUSTAL O(1.2.4) multiple sequence alignment


Danio      MSLSDTDKAVVKAIWAKISPKADEIGAEALARMLTVYPQTKTYFSHWADLSPGSGPVKKH 60
Pan        MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKGH 59
Homo       MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF-DLSHGSAQVKGH 59
Mus        MVLSGEDKSNIKAAWGKIGGHGAEYGAEALERMFASFPTTKTYFPHF-DVSHGSAQVKGH 59
Capra      MSLTRTERTIILSLWSKISTQADVIGTETLERLFSCYPQAKTYFPHF-DLHSGSAQLRAH 59
           * *:  ::: : : *.*:. :.   *:*:* *::  :* :**** *: *:  **. :: *

Danio      GKTIMGAVGEAISKIDDLVGGLAALSELHAFKLRVDPANFKILSHNVIVVIAMLFPADFT 120
Pan        GKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPAEFT 119
Homo       GKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPAEFT 119
Mus        GKKVADALANAAGHLDDLPGALSALSDLHAHKLRVDPVNFKLLSHCLLVTLASHHPADFT 119
Capra      GSKVVAAVGDAVKSIDNVTSALSKLSELHAYVLRVDPVNFKFLSHCLLVTLASHFPADFT 119
           *..:  *: :*   :*:: ..*: **:***. *****.***:*** ::*.:*   **:**

Danio      PEVHVSVDKFFNNLALALSEKYR      143
Pan        PAVHASLDKFLASVSTVLTSKYR      142
Homo       PAVHASLDKFLASVSTVLTSKYR      142
Mus        PAVHASLDKFLASVSTVLTSKYR      142
Capra      ADAHAAWDKFLSIVSGVLTEKYR      142
             .*.: ***:  :: .*:.***
```

5. We will be able to establish evolutionary relationships between the sequences, for this we can explore the “Guide tree” button. We must remember that Clustal Omega is only a multiple sequence alignment program, it is not a phylogenetic analysis program.

6. Click the “Result Files” tab, then go to “Percent Identity Matrix” and check for the table that shows the percentage of identical residues between pairs of sequences in this multiple sequence alignment.

```
Percent Identity  Matrix - created by Clustal2.1


  1: Danio       100.00   53.52   53.52   54.23   49.30
  2: Pan          53.52  100.00  100.00   86.62   56.34
  3: Homo         53.52  100.00  100.00   86.62   56.34
  4: Mus          54.23   86.62   86.62  100.00   56.34
  5: Capra        49.30   56.34   56.34   56.34  100.00
```

Answer in class:

Q1. Could we consider the Tsol18 and Tso45 sequences as homologous?

- Accession number of the sequences consulted:
- Substitution matrix used by the algorithm:
- Shows a screenshot of the alignment:
- Alignment score:
- Identity percentage:
- Similarity percentage:
- Number of gaps:
- Orthologous sequences are a type of homologous sequences that at some point in evolutionary history were separated by a speciation event. Could it be concluded that the sequences analyzed are orthologous to each other? Why?

Q2. Analyze the guide tree and answer, do the proposed relationships make any biological sense?